

AWS S3 Static Website Hosting

Objective

The objective of this task was to create an Amazon S3 bucket, configure it for static website hosting, upload a simple `index.html` file, configure the required public access permissions, and verify the website using the S3 website endpoint.

Description

1. Create a S3 bucket inside your aws instace.
2. change the properties of bucket so that it allows website hosting.
3. add a simple static website to the S3 created.
Note: try with a simple `index.html` file.
4. check the required permission/access control of the bucket.
5. verify the working using the endpoint of bucket.

1. Create the S3 Bucket

An Amazon S3 bucket named **s3-webhosting123** was created in the **Asia Pacific (Mumbai) – ap-south-1** region.

During bucket creation, the required settings were configured for the static website hosting lab. Object Ownership was kept as **Bucket owner enforced**, which disables ACLs and allows access to be controlled using bucket policies.

Block Public Access was disabled for the bucket because the static website needed to be publicly accessible through the S3 website endpoint.

Versioning was not enabled, and the default S3 server-side encryption using **SSE-S3** was retained.

The bucket was successfully created and was ready for website configuration.

2. Enable Static Website Hosting

The properties of the **s3-webhosting123** bucket were opened and the **Static website hosting** option was enabled.

The hosting type was configured as:

Host a static website

The index document was specified as:

index.html

After saving the configuration, Amazon S3 generated a static website endpoint for the bucket.

The website endpoint was:

<http://s3-webhosting123.s3-website.ap-south-1.amazonaws.com/>

This endpoint was later used to verify that the website was working.

3. Create and Upload the Static Website

A simple HTML file named **index.html** was created for the static website.

The contents of the file were:

```
<!DOCTYPE html>
<html>
<head>
  <title>My S3 Website</title>
</head>
<body>
  <h1>Welcome to My S3 Static Website</h1>
  <p>This website is hosted using Amazon S3.</p>
</body>
</html>
```

The **index.html** file was uploaded to the root of the **s3-webhosting123** S3 bucket.

The file was successfully uploaded and was available as an object inside the bucket.

4. Configure Permissions and Access Control

The bucket's public access settings were checked to allow the static website to be accessed through the S3 website endpoint.

A bucket policy was added to allow public read access to objects inside the bucket.

The policy used was:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadForWebsite",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::s3-webhosting123/*"
    }
  ]
}
```

The policy allowed users to perform:

`s3:GetObject`

on objects inside the bucket.

This permission was required so that visitors could retrieve the `index.html` file through the website endpoint.

The policy did not provide permissions to upload, modify, or delete objects.

5. Verify the Working Using the Endpoint

The S3 static website endpoint was opened in a web browser:

<http://s3-webhosting123.s3-website.ap-south-1.amazonaws.com/>

The website loaded successfully and displayed:

Welcome to My S3 Static Website

and:

This website is hosted using Amazon S3.

This confirmed that the `index.html` file was being served successfully through the S3 static website endpoint.

The browser displayed **Not secure** because the S3 static website endpoint uses HTTP rather than HTTPS.

Screenshot



Welcome to My S3 Static Website

This website is hosted using Amazon S3.

Conclusion

The S3 static website hosting task was successfully completed. An S3 bucket named **s3-webhosting123** was created in the `ap-south-1` region, static website hosting was enabled, and a simple `index.html` webpage was uploaded.

The required public read permission was configured using an S3 bucket policy. Finally, the generated S3 website endpoint was accessed through a web browser, and the webpage was successfully displayed.